Jupyter Cheat Sheet · JupyterLab and Notebook

QuantStack · Jupyter Explained · No hype. Just code. · Shortcuts and menu names change between versions: check them on yours.

LAUNCH AND STOP

- `jupyter lab` · `jupyter notebook`
- `jupyter server list` shows running servers
- Ctrl+C in the prompt stops the server
- **Close and Shut Down Notebook** stops one notebook's kernel
- **File → Shut Down** stops the whole server
- Closing a tab does not stop the kernel
- Jupyter shows the folder you launched it from

ENVIRONMENTS (CONDA)

- `conda env list` · `conda list`
- `conda create -n NAME python=3.xx`
- `conda activate NAME` · `conda deactivate`
- `conda env remove -n NAME`
- `conda install ipykernel` in the environment
- `python -m ipykernel install --user --name NAME --display-name "Python (NAME)"`
- `jupyter kernelspec list` · `jupyter kernelspec remove NAME`

INSTALLING PACKAGES

- Conda first, pip last, don't alternate
- `python -m pip install PKG` uses this Python's pip
- In a notebook: `%pip install PKG` · `%conda install PKG`
- Avoid `!pip install`: it can hit another Python
- `conda env export --from-history > environment.yml`
- `conda env create -f environment.yml`
- ModuleNotFoundError? Ask which Python the kernel runs: `import sys; sys.executable`

CELLS (COMMAND MODE)

- **Esc** command mode · **Enter** edit mode
- **A** insert above · **B** insert below
- **D D** delete · **Z** undo delete
- **C** copy · **X** cut · **V** paste below
- **M** markdown · **Y** code
- **Shift+↑/↓** select · **Shift+M** merge
- **Ctrl+Shift+-** split a cell (edit mode)
- **J / K** or arrows move between cells · **Shift+V** paste above

RUN AND SAVE

- **Shift+Enter** run, move on
- **Ctrl+Enter** run, stay
- **Alt+Enter** run, add a cell below
- **I I** interrupt · **0 0** restart
- **Ctrl+S** save
- Counter: empty = not run · * = running · number = run order
- Higher number above a lower one = out of order

JUPYTERLAB

- **Ctrl+Shift+C** command palette: search any command
- **Ctrl+B** show or hide the left sidebar
- **Ctrl+F** search the notebook
- **Tab** completes · **Shift+Tab** shows help
- `name?` docs · `name??` source
- Drag tabs to split the screen
- **Ctrl+/** comment lines (edit mode)
- Settings → Settings Editor → shortcuts
- Stuck layout? add `?reset` to the address (save first)

KERNEL MENU

- **Interrupt**: stops a stuck cell, keeps your variables
- **Restart**: forgets everything
- **Restart and Clear Outputs**: use before sharing
- **Restart and Run All**: the honest test
- **Change Kernel**: switching restarts it
- Run → **Run All Above** before a dependent cell

MAGICS AND SHELL

- `%timeit` line · `%%timeit` cell · `%%time` once
- `%who` · `%whos` · `%history` · `%lsmagic`
- `%env` can show secrets: blur it
- `%%writefile name.py` (first line) · `%run name.py` · `%load name.py`
- `%debug` after an error · `q` quits
- `%pwd` · `%ls` · `%cd` · `%matplotlib inline` · `%%HTML`
- `%load_ext autoreload` then `%autoreload 2`
- _ last output · `Out[n]` · `In[n]` · `!command` runs in the shell

EXPORT AND SAFETY

- `jupyter nbconvert --to html nb.ipynb` · `--to markdown` · `--to script` · `--no-input`
- `jupyter nbconvert --execute --to notebook --output out.ipynb nb.ipynb`
- Before sharing: restart and run all, clear or keep outputs, scan for tokens and paths
- Never expose a server · never remove the token
- `jupyter server password` · `jupyter --paths`
- Read a notebook's code before running someone else's

When something breaks

Symptom, likely cause, fix, and the part of the series that explains it. Check commands and menu names on your version.

SYMPTOM	LIKELY CAUSE	FIX	PART
`'jupyter' is not recognised`	Wrong prompt (Windows)	Open Anaconda Prompt	1
Browser didn't open	Server is running, tab wasn't launched	Copy the full address, token included	1
Can't find my notebook	Launched from another folder	`cd` first, then `jupyter lab`; check the file browser's root	1
Port already in use	Another server is running	`jupyter server list`, stop it, or use `--port`	1
`ModuleNotFoundError`	Package is in another Python	`import sys; sys.executable`, then `%pip install` or conda in that environment	2
Kernel won't start	Its environment was deleted	Remove the stale kernel, pick another, or rebuild the environment	2
Stuck layout or odd tabs	Saved workspace	Add `?reset` to the address (save first)	3
Output doesn't match the code	Cells ran out of order	Restart and Run All; check the counters	4
Cell shows `[*]` forever	Still running or stuck	Kernel → Interrupt; if that fails, Restart	4
Kernel died	Often out of memory	Restart, load less data, delete what you don't need	4
Edited module has no effect	Notebook kept the old import	`%load_ext autoreload` then `%autoreload 2`, or restart	5
Export to PDF fails	Needs extra software	Export to HTML, then print to PDF from the browser	6
Shared a file with a secret in it	Outputs are saved in the file	Rotate the secret, clear outputs, scan before sharing	6